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Autospores (spores which look similar to parent cell Example: *Chlorella*); Hypnospore (thick walled aplanospore – Example: *Chlamydomonas nivalis*); Tetraspores (Diploid thallus of *Polysiphonia* produce haploid spores after meiosis).

Sexual reproduction in algae are of three types 1. Isogamy (Fusion of morphologically and Physiologically similar gametes Example: *Ulothrix*) 2. Anisogamy (Fusion of either morphologically or physiologically dissimilar gametes Example: *Pandorina*) 3. Oogamy (Fusion of both morphologically and physiologically dissimilar gametes. Example: *Sargassum*). The life cycle shows distinct alternation of generation.



The Oldest recorded alga is Grypania, which was discovered in the banded iron formations of northern Michigan and dated to approximately 2100Ma

2.3.2. Classification

F.E. Fritsch proposed a classification for algae based on pigmentation, types of flagella, reserve food materials, thallus

Table 2.2 Classification of Algae

Class	Pigments	Flagella	Reserve food
Chlorophyceae	Chlorophyll a and b Carotenoids, Xathophyll	1,2,4 or more equal anterior whiplash flagella	Starch
Xanthophyceae	Chlorophyll a and b Carotenoids Xathophyll	2, unequal anterior 1 tinsel and 1 whiplash	Fats and leucosin
Chrysophyceae	Chlorophyll a and b Carotenoids,	1 or 2 unequal or equal anterior both whiplash or 1 whiplash and 1 tinsel	Oils and leucosin
Bacillariophyceae	Chlorophyll a and c Carotenoids,	1 anterior (only in male gametes) tinsel	Leucosin and Fats
Cryptophyceae	Chlorophyll a and c carotenoids and xanthophyll	unequal anterior both tinsel flagella	Starch
Dinophyceae	Chlorophyll a and c carotenoids and xanthophyll	Two unequal (whiplash) lateral flagella in different plane	Starch and oil
Chloromonadineae	Chlorophyll a and b Carotenoids, Xathophyll	2 equal flagella	oil
Euglenophyceae	Chlorophyll a and b	One or two anterior tinsel flagella	Fats and paramylon
Phaeophyceae	Chlorophyll a and c, Xanthophyll	Two unequal whiplash and tinsel lateral flagella	Laminarin starch and fats
Rhodophyceae	Chlorophyll a, r-Phycoerthythrin	absent	Floridean starch
Cyanophyceae	Chlorophyll a, carotenoids, c-Phycocyanin,Allophycocyanin	absent	Cyanophycean starch